

AI Quiz Prep Reference Map

This 1-page cheat sheet links course lectures, survey papers, and hands-on repos/tools. Use it to anchor quiz answers by combining **course concepts** with **modern references** and **evaluation frameworks**.

Course Concept	Lecture / Slide	Survey / Research	Repos / Tools	Quiz Use
Data Types, Prep, Ontologies	Weeks 1–3 (Intro, Data Representation)	Prompt Report – Input formats; structured vs. unstructured prompts	–	Define structured/unstructured data; link ontologies to KBs.
Supervised / Unsupervised ML	Weeks 4–6 (Clustering, Neural Nets, Text classification)	Prompt Report – Few-shot vs. zero-shot prompting	AI Suite (Andrew Ng)	Explain classification vs. prompting strategies.
Trust, Explainability, Rating	Week 6 (Trust & Logic)	Prompt Report – evaluation frameworks; Model Collapse (Nature, 2024)	Promptify (evaluation chains)	Trade off explainability vs. accuracy; trust rating frameworks.
Logic & Reasoning (FOL, Semi-decidable)	Week 7 Lecture 13	Prompt Report – prompt chaining & answer engineering	Promptify (template chaining)	Show inference pitfalls; decidable vs. semi-decidable.
Search (BFS, DFS, UCS, IDS, Bidirectional)	Week 7 Lecture 14	Prompt Report – rubric/answer engineering	Notebook + Promptify	Compare strategies; analyze LLM consistency.
LLMs (Transformers, Encoders/Decoders, Prompting)	Weeks 5–6 (LLMs, Transformer lecture)	Prompt Report – taxonomy of prompting	Promptify + AI Suite	Explain architectures; demonstrate prompting impacts.
Prompting Basics (4-part structure)	Week 5 Lecture 10	Prompt Report – 58 techniques (zero/few-shot, CoT, role)	Promptify (prompt templates)	Write prompts with instruction/context/input/output + rubric.
Evaluation of Outputs (Stability, GAICO)	Weeks 6–7 Exercises	Prompt Report – metrics (accuracy, diversity, faithfulness)	GAICO + Notebook	Compare outputs; stability across reruns; rubric coverage.
Risks in AI (Data, Sustainability)	General + W6 invited talk	Model Collapse (Nature, 2024)	–	Explain recursive training risks; need human data.

Tip. If asked a conceptual question, cite **Lectures** first. If asked an application or GenAI exercise, reference **Repos/Tools** and evaluation workflow. For prompting, cite both Lecture 10 (4-part structure) and *Prompt Report*.